



DAT-210-01
Data Analytics
FINAL PROJECT

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Abstract

This project analyzes how different factors influence individual income using the Adult dataset. We examined multiple variables including age, education, occupation, race, gender, and work hours to identify patterns and determinants of potential income. We implemented three classification algorithms (Support Vector Machine, Decision Tree, and Random Forest) to predict whether an individual's annual income exceeds \$50,000. Model analysis determined that the Random Forest classifier outperformed the others, having a higher precision and F1 score. This research contributes to understanding how different factors impact income as well as the reliability of these models in similar datasets.

1 Introduction

We set out to interpret the features and structure of the adult dataset. We started with an initial data analysis to gain a basic understanding of the dataset, from which we were able to derive our own questions to extract even more insights:

- How does age affect income? What age earns the most?
- How do hours-per-week relate to education level? Does a lower level of education mean higher hours per week?
- What are the most common features of the dataset and what percentage of the dataset do people with the features take up?
- What is the average capital (gain & loss) for each occupation?
- Is there a difference in occupation between people native to the United States and people who are not?

Armed with this information, we set out to build and evaluate classification models using the Support Vector Machine, Decision Tree, and Random Forest models to gain an even further understanding of the dataset's core metrics.

2 Data Explanation & Initial Analysis

The dataset is collected from the 1994 US census and contains fifteen attributes for each of the 45222 entries. Attribute labels and statistics are included in the tables below.

	workclass	education	marital-status	occupation	relationship	race	gender	native-country	income
count	45222	45222	45222	45222	45222	45222	45222	45222	45222
unique	7	16	7	14	6	5	2	41	2
top	Private	HS-grad	Married-civ-spouse	Craft-repair	Husband	White	Male	United-States	<=50K
freq	33307	14783	21055	6020	18666	38903	30527	41292	34014

Figure 1: Statistics for categorical data

Of note, some of the attributes (work class, race, and native country) have one category occurring more frequently than all other categories combined despite having five or more values. Native country in particular has 91% of all entries as United-States. Other countries can have so few entries that they are not useful for data analysis. These countries should be bundled into a single category.

	age	fnlwgt	educational-num	capital-gain	capital-loss	hours-per-week
count	45222.000000	4.522200e+04	45222.000000	45222.000000	45222.000000	45222.000000
mean	38.547941	1.897347e+05	10.118460	1101.430344	88.595418	40.938017
std	13.217870	1.056392e+05	2.552881	7506.430084	404.956092	12.007508
min	17.000000	1.349200e+04	1.000000	0.000000	0.000000	1.000000
25%	28.000000	1.173882e+05	9.000000	0.000000	0.000000	40.000000
50%	37.000000	1.783160e+05	10.000000	0.000000	0.000000	40.000000
75%	47.000000	2.379260e+05	13.000000	0.000000	0.000000	45.000000
max	90.000000	1.490400e+06	16.000000	99999.000000	4356.000000	99.000000

Figure 2: Descriptive statistics of numerical data

The capital gain and loss quartiles are all zero. This will heavily skew the data and make capital gain and loss difficult. The max capital gain is also suspect with a value of 99999. Considering the next highest capital gains value is 41310, the 99999 is likely to be representative of a null value. There are 229 entries of 99999 that should be excluded from analysis.

	age	fnlwgt	educational-num	capital-gain	capital-loss	hours-per-week
age	1.000000	-0.075792	0.037623	0.079683	0.059351	0.101992
fnlwgt	-0.075792	1.000000	-0.041993	-0.004110	-0.004349	-0.018679
educational-num	0.037623	-0.041993	1.000000	0.126907	0.081711	0.146206
capital-gain	0.079683	-0.004110	0.126907	1.000000	-0.032102	0.083880
capital-loss	0.059351	-0.004349	0.081711	-0.032102	1.000000	0.054195
hours-per-week	0.101992	-0.018679	0.146206	0.083880	0.054195	1.000000
over_50k	0.237040	-0.007264	0.332800	0.221034	0.148687	0.227199

Figure 3: correlation table of numerical data

There is little correlation between the numeric attributes.

3 Explanatory Data Analytics

3.1 How does age affect income? What age earns the most?

Due to the income variable containing a binary value, further analysis requires a different metric for evaluation based on this. For this analysis we tallied how many people earned above 50k and divided by the total people in the dataset and multiplied by 100. To derive the percentage of people earning over 50k. After finding the percentage of people earning over 50k for each age we noticed that some ages were outliers due to low sample size. To remedy this we grouped the ages by age bracket to account for low sample size at the ends of the age range. The individual age with the highest earnings is 50.



Figure 4: Percentage of people earning over 50k for each age

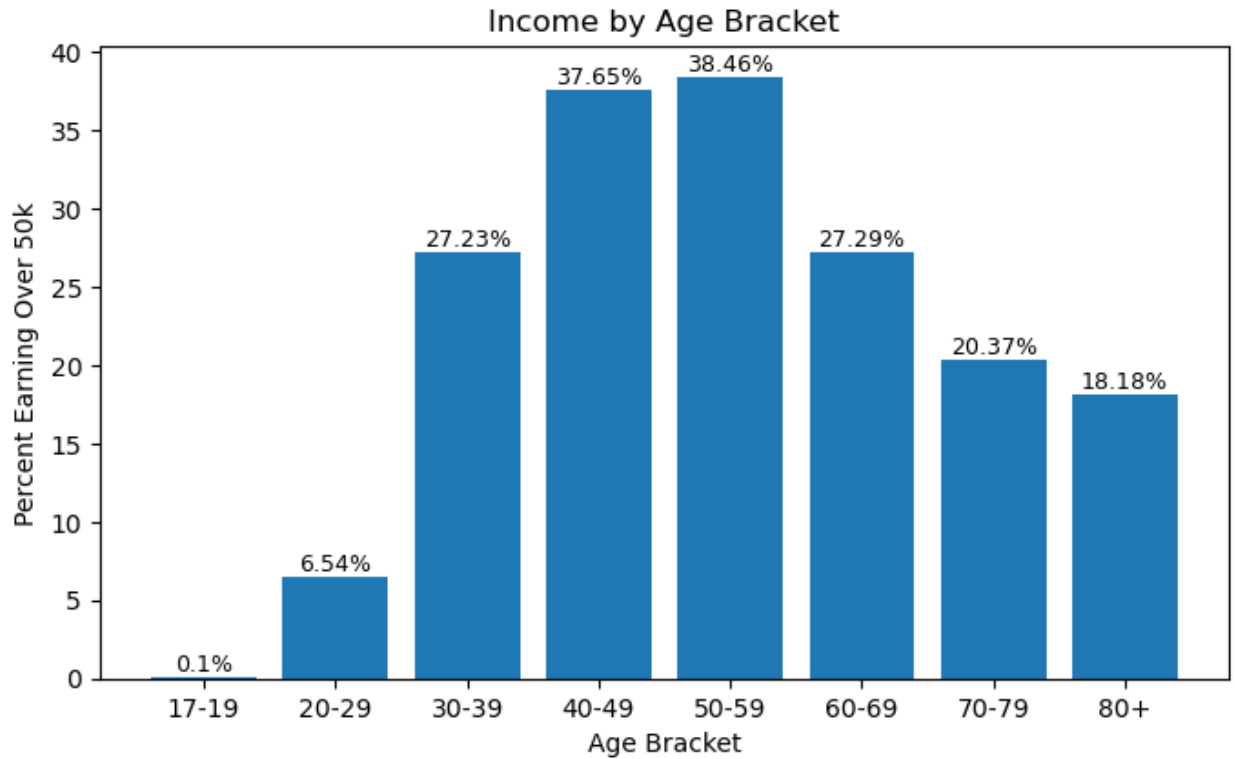


Figure 5: Percentage of people earning over 50k grouped by age bracket

3.2 How do hours-per-week relate to education level? Does a lower level of education mean higher hours per week?

The original hypothesis for this question would be that lower education would lead to a lower paying job. A lower paying job would mean that someone would need to work more hours in order to be self-sufficient. Interestingly, the results from the data shows the opposite — People with a higher education tend to work more hours on average. There could be a variety of reasons for this, such as people that are able for afford a higher education are able to afford to spend more time at work.

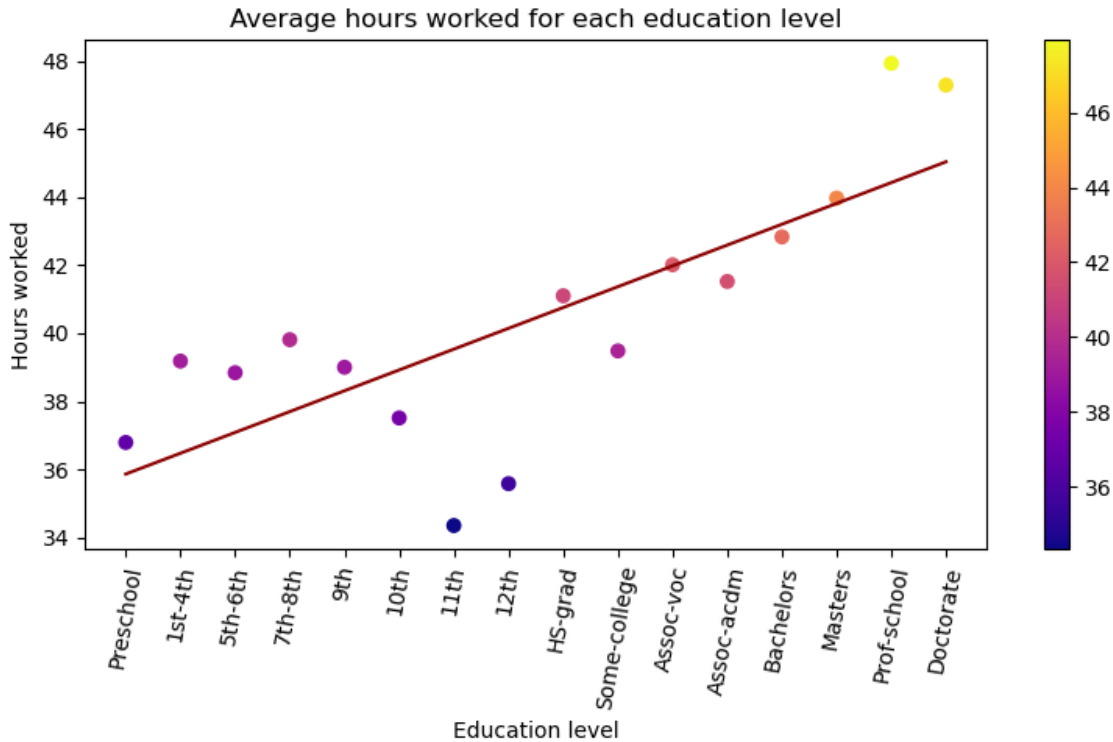


Figure 6: Average — Hours worked for each education level

3.3 What are the most common features of the dataset and what percentage of the dataset do people with the features take up?

For this we sorted the data by final weight (fnlwgt column in the dataset¹). Final weight is the number of people the census believes the entry represents. After sorting, we can see that the most common aspects are:

- Private work class
- No spouse (divorced/separated/spouse absent)
- White
- From the US
- Less than \$50k income

Our end goal is to see what percentage do people with the most common features make up in the dataset. From the data we were able to derive that 31% of the dataset shares all of these features.

What percentage of the dataset shares the most common features

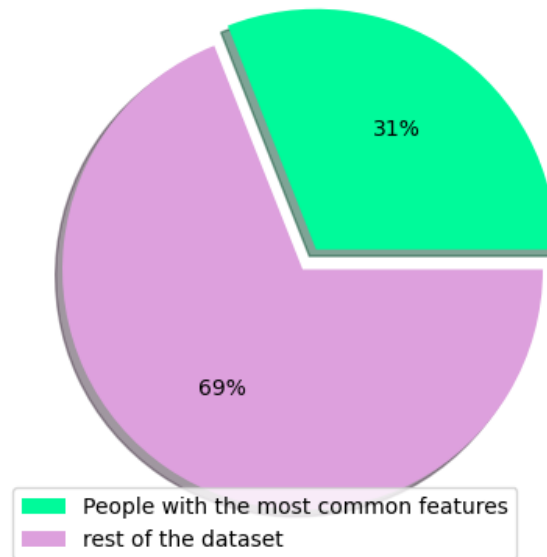


Figure 7: Percentage — People with the most common features in the dataset

3.4 What is the average capital for each occupation?

We took both capital gain and capital loss into account when researching the average capital for each occupation. Due to the data set reporting both columns as positive numbers, we had to take this into account and enter the capital loss as a negative number into our calculations. The occupation with the highest average capital is the Exec-Managerial occupation.

¹Information about fnlwgt via Becker & Kohavi and UCI Machine learning sources

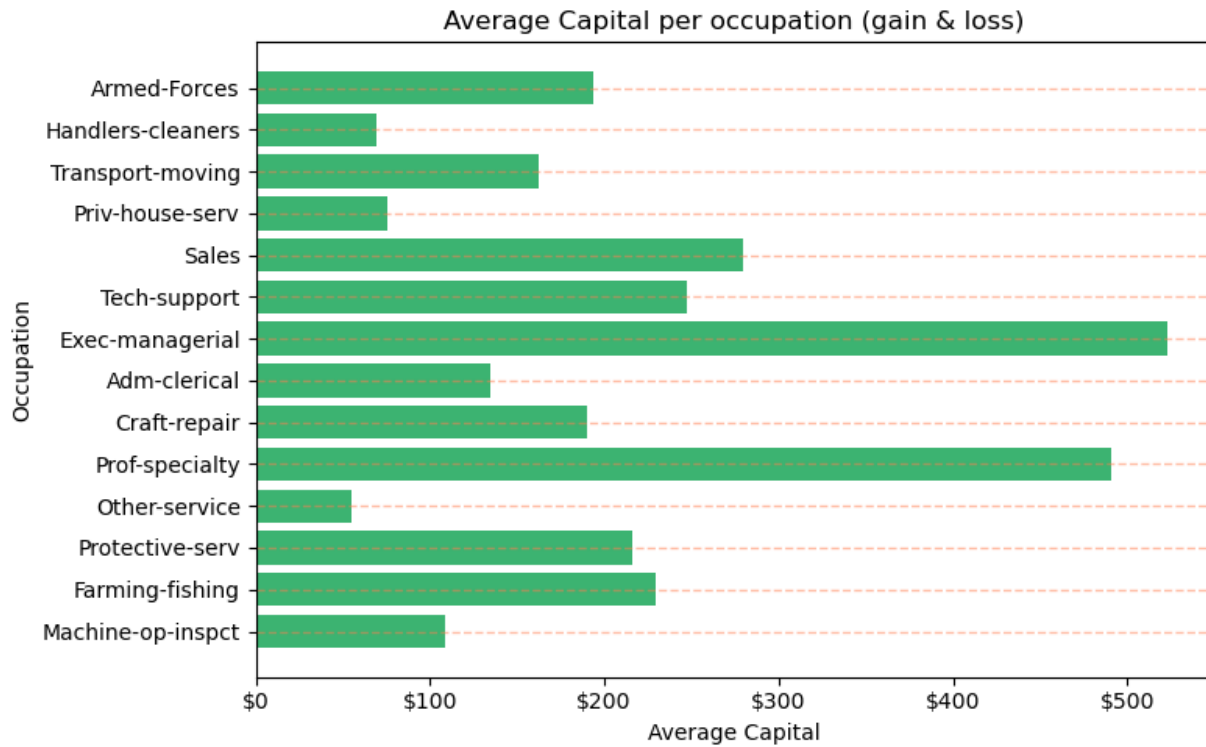


Figure 8: Average — Capital per occupation

3.5 Is there a difference in occupation between people native to the United States and people who are not?

We split the data by the native-country variable and whether or not it was equal to 'United-States'. We found the percentage of people with an occupation for every occupation in both sets and plotted the results. The most popular job for US natives was exec-managerial, but the same job was not as popular among non-natives.

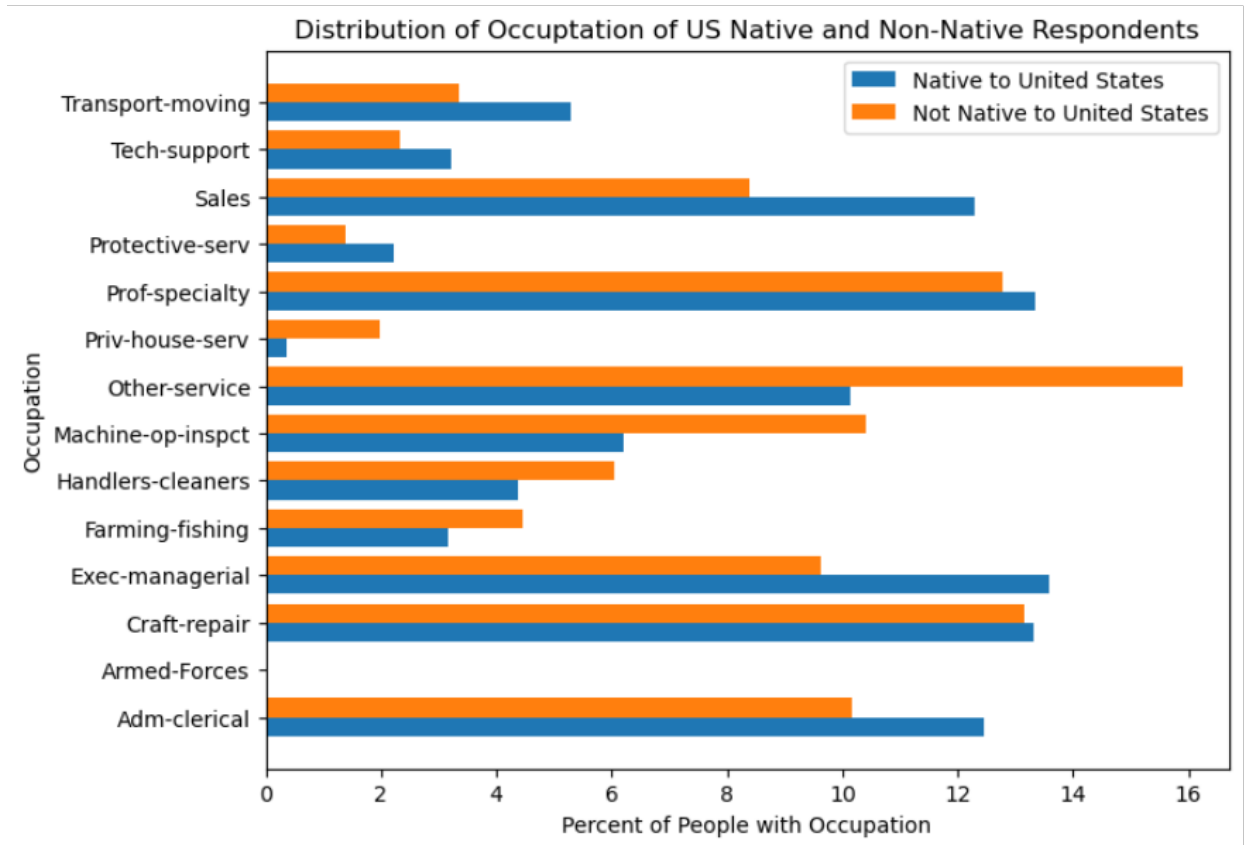


Figure 9: Occupation distribution of US natives and non-natives

4 Classification & Model Evaluation

We used 3 models for evaluating the dataset, and we scored each model on accuracy, precision, recall, and F1 score. The first model we made and tested is the Support Vector Machine. A Support Vector Machine is a supervised machine learning algorithm used for classification and regression tasks. While it can handle regression problems, SVM is particularly well-suited for classification tasks. A Support Vector Machine aims to find the optimal hyperplane in an N-dimensional space to separate data points into different classes. The algorithm maximizes the margin between the closest points of different classes. The next model we used is a Decision Tree. A Decision Tree model — similar to a Support Vector Machine — is a supervised learning algorithm used for both

classification and regression tasks. It models decisions as a tree-like structure where internal nodes represent attribute tests, branches represent attribute values and leaf nodes represent final decisions or predictions. Decision trees are versatile, interpretable, and widely used in machine learning for predictive modeling. The third and final model we used is a Random Forest. The Random Forest model had similarities to the Decision Trees model which is due to the fact that a Random Forest is a collection of decision trees that work together. It takes different random parts of the dataset to train each tree and then it combines the results by averaging them. This approach helps improve the accuracy of predictions.

	Accuracy	Precision	Recall	F1 Score
Support Vector Machine	0.84212	0.75745	0.56128	0.64478
Decision Tree	0.81305	0.63296	0.63707	0.63501
Random Forest	0.84854	0.74264	0.62235	0.67719

Table 1: Classification Metrics Comparison

The data is split into 4 sections: Accuracy, Precision, Recall, and F1 Score. Accuracy is essentially "Out of all the predictions we made, how many were true?".¹

$$Accuracy = \frac{\text{true positives} + \text{true negatives}}{\text{true positives} + \text{true negatives} + \text{false negatives} + \text{false positives}}$$

Precision is a metric that gives the proportion of true positives to the amount of total positives that the model predicts. Summarized: "Out of all the positive predictions we made, how many were true?"¹

$$Precision = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

Recall is focused on finding out how good a model is at finding all of the positives. Summarized as "Out of all the data points that should be predicted as true, how many did we correctly predict as true?"¹

$$Precision = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

F1 score is a measure that combines the previous 2 categories, recall and precision. Due to there being a trade off between the 2 categories, F1 score measures how the model effectively interacts with the trade off.

$$F1Score = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The prioritization of different metrics would depend on a few different factors in the dataset. In the Adult Dataset the most important of these would be F1 score, as the data is imbalanced — People earning more than \$50k and less isn't a 50/50 split. Another thing to account is that in this dataset both false positives and false negatives have meaningful consequences if the data is used elsewhere, such as financial assistance being denied if a service is using a different metric. F1 score provides a balanced view that accuracy, precision, and recall can provide by themselves. Below is a simple graph comparing how all 3 models compare to one another in relation to these 4 metrics:

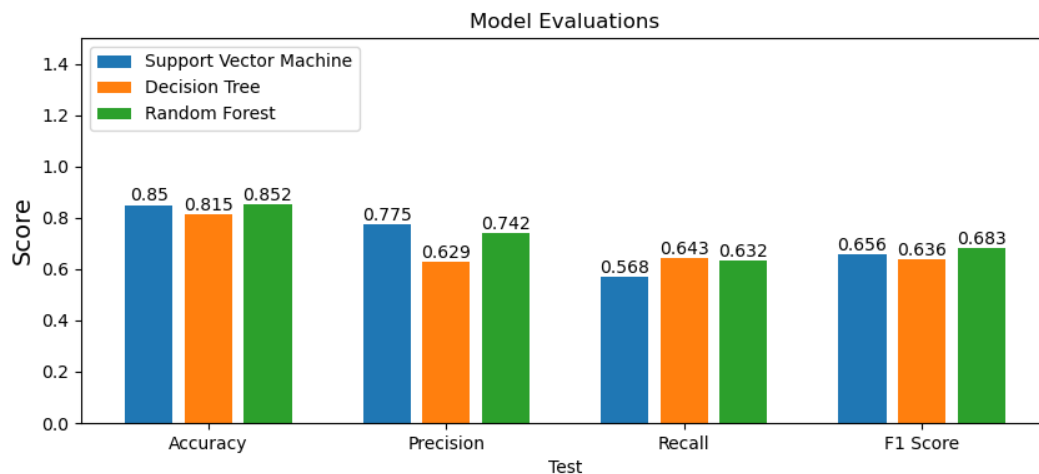


Figure 10: Evaluation of all 3 models

5 Individual Contributions

We divided the project into equal parts. For Sections 2.2 and 2.3 we both answered five questions each. In section 2.2 Patrick answered questions 1, 3 and 5 meanwhile Conor answered questions 2 and 4. In section 2.3 Conor posed and answered 3 advanced analytical questions — questions 2, 3 and 4 — and Patrick did questions 1 and 5. For section 2.4 — Classification and Model Evaluation — We both worked on the models together because we both needed to use the models and their data. Throughout the project we divided the work as evenly as possible.

¹Information about metrics via Tigerschiold

5.1 Person Reflections

5.1.1 Patrick's Reflection

Throughout this project, I learned to ask the right questions about datasets and to understand what is and is not relevant to a given task. I struggled with this initially, but eventually found interesting ways to investigate the data. I also learned how to show my findings for these questions in a way that is clear and easy to understand. For my work process, I should have worked on this project in smaller increments rather than working 3+ hours at a time. This would have made work less stressful and more efficient.

5.1.2 Conor's Reflection

Overall I have enjoyed this project as it made me think "outside the box" when it comes to asking analytical questions. While I did struggle with it at the beginning with figuring out questions that would be considered "advanced analytical questions" I found the best solution to think about a wild connection to the data with a "what if..." and see if I could code a table that both looked good and could be considered advanced. This project also used elements that we have learned over the semester which I liked as I felt completely prepared.

6 Conclusion and Recommendations

From our findings we are able to see how some aspects directly relate to income and indirectly. Starting off with section 2.2: How does income vary across education levels? Overall the higher an education someone has the higher their income, with the group having the highest percentage earning over \$50k being individuals with masters degrees. What is the frequency distribution of income levels for each work class category? A common theme in each category is that majority of the dataset earn less than \$50k, with majority of the dataset being in the private work class. How does the average age differ between individuals earning greater than \$50k and those earning less than or equal to \$50k? On average people earning greater than \$50k are 7.26 years older than people earning less than or equal to \$50k. The average age of people earning less than \$50k is 36.75 and the average for people earning above \$50k is 44.01. What is the relationship between marital status and income levels? From our analysis we found that 44.51% of married adults have an income above \$50k and 6.68% having an income less than \$50k. Is there a significant difference in capital gains between high-income and low-income groups? In our reasearch we saw that because the majority of people earn no capital gains there is a significant gap between the two groups. The average capital-gains of people earning less than or equal to \$50k is \$149.02 and the average capital-gains of people earning greater than \$50k is \$3991.79.

In section 2.3 we posed our own advanced analytical questions to answer. How does age impact income? What age earns the most on average? What we found is that income is distributed in a bell curve with the highest point being the 50-59 age bracket. How do hours-per-week relate to education level? Does a lower level of education mean higher hours per week? Overall we found that a higher education level lead to working more hours per week on average. What are the most common features of the dataset and what percentage of the dataset do people with the features take up? After sorting the dataset by final weight we found the most common features to be: private work class, no spouse (divorced/separated/spouse absent), white, from the US, and less than \$50k income. Throughout the entire dataset individuals with these features took up 31% of the data. What is the average capital for each occupation? We found that because most individuals in the dataset had very little to no capital gain and loss the averages to be relatively low, regardless the occupation with the highest capital on average is Exec-Managerial occupation. Is there a difference in occupation between people native to the United States and people who are not? We found that choice of occupation varies significantly between people native to the United States and people who are not.

Building and training 3 different models to look at the dataset and predict if an individuals income is greater than \$50k lead to us finding that on average the Random Forest model to be the best out of the 3. The Random forest scored the highest followed by the Support Vector Machine model.

Going forward it would be best when analyzing similar datasets — in relation to income — to use the Random Forest model as it scored the highest in F1 score compared to Support Vector Machine and Decision Tree models as F1 score was the metric we decided was the most important.

7 Works cited

Becker, Barry, and Ronny Kohavi. “UCI Machine Learning Repository.” Archive.ics.uci.edu, 30 Apr. 1996, archive.ics.uci.edu/dataset/2/adult.

Tigerschiold, Ted. “What Is Accuracy, Precision, Recall and F1 Score?” Www.labelf.ai, 7 Nov. 2022, www.labelf.ai/blog/what-is-accuracy-precision-recall-and-f1-score.

UCI Machine Learning, and Kaggle Team. “Adult Census Income.” Www.kaggle.com, 2014, www.kaggle.com/datasets/uciml/adult-census-income.